

Telco Customer Churn Analysis

1. Project Overview

Introduction

The Telco Customer Churn Analysis project aims to analyse customer data to identify the key factors affecting customer churn and revenue. Using Excel, Python (Pandas), PostgreSQL, and Power BI, the project transforms raw data into meaningful insights through data cleaning, SQL analysis, KPI creation, and interactive dashboards.

The analysis focuses on customer behaviour, contract types, payment methods, internet services, and revenue trends to understand why customers leave and which customer segments generate the highest value. The insights help businesses make data-driven decisions to improve customer retention, increase profitability, and enhance overall business performance.

Objective

- Analyse customer churn in a telecom company.
- identify the factors responsible for customer attrition.
- provide actionable insights through interactive Power BI dashboards.
- helps business stakeholders understand customer behaviour.
- revenue contribution, contract preferences, payment methods.
- service usage to reduce churn and improve customer retention.

2. Problem Statement

Customer churn is one of the biggest challenges faced by telecom companies. Losing existing customers directly impacts revenue and business growth.

The company wants answers to questions like:

- Why are customers leaving?
- Which customers are at higher risk?
- Which services contribute to churn?
- Which contract type generates maximum revenue?
- Which payment methods are preferred?
- Which customers generate the highest revenue?

This project answers all these business questions using data analysis.

3. Business Objectives

- Analyse customer demographics.

- Measure customer churn.
- Identify high-risk customers.
- Understand customer behaviour.
- Analyse revenue generation.
- Improve customer retention.
- Support business decision-making.
- Build an interactive reporting dashboard.

4. Dataset Information

Attribute	Details
Dataset Name	Telco Customer Churn
Total Records	7,043 Customers
Columns	21
Format	CSV
Domain	Telecom
Missing Values	Total_charges
Target Column	Churn

5. Tools & Technologies Used

Tool	Purpose
Power BI	Dashboard Creation & Data Visualization
PostgreSQL	Data Storage & SQL Analysis
Python	Data Cleaning & Data Processing
Pandas	Data Manipulation & Analysis
Excel	Initial Data Cleaning & Validation

6. Project Architecture

Data Collection (CSV Dataset)



Data Cleaning (Excel & Python)



Data Transformation (Pandas)



Data Storage (PostgreSQL)



SQL Analysis & Business Queries



Data Modelling in Power BI



DAX Measures & KPIs



Interactive Dashboards



Business Insights & Recommendations

7. Data Collection

The dataset contains customer information including:

- Customer ID
- Gender
- Senior Citizen
- Partner
- Dependents
- Tenure
- Phone Service
- Internet Service
- Online Security
- Tech Support

- Streaming TV
- Streaming Movies
- Contract
- Payment Method
- Monthly Charges
- Total Charges
- Churn

8. Data Cleaning

Data cleaning was performed using **Excel** and **Python (Pandas)**.

Cleaning Steps

- Removed duplicate records
- Checked missing values
- Converted data types
- Removed blank spaces
- Replaced null values
- Standardized categorical values
- Converted Total Charges into numeric format
- Verified data consistency

9. Data Preprocessing

Python libraries used

Pandas

NumPy

Operations performed

- Import dataset
- Handle missing values
- Convert datatypes
- Remove duplicates
- Export cleaned dataset

10. PostgreSQL Analysis

The cleaned dataset was imported into PostgreSQL.

SQL operations performed:

- SELECT
- WHERE
- GROUP BY
- ORDER BY
- COUNT()
- SUM()
- AVG()
- CASE
- Window Functions
- Aggregate Functions

More than **80 SQL queries** were executed to analyze customer behavior and business performance.

11. Power BI Development

Power BI was used to create three interactive dashboards.

Dashboard 1

Executive Overview

Provide an overall business summary.

KPIs

- Total Customers
- Active Customers
- Churned Customers
- Churn Rate
- Total Revenue
- Average Monthly Charges
- Average Tenure

Charts

- Customers by Gender
- Customers by Contract
- Customers by Internet Service
- Revenue by Contract
- Customers by Payment Method
- Monthly Charges by Contract

Dashboard 2

Customer Churn Analysis

Identify factors responsible for customer churn.

KPIs

- Total Churned Customers
- Churn Rate
- Average Monthly Charges (Churned)
- Average Tenure (Churned)

Charts

- Churn by Contract Type
- Churn by Internet Service
- Churn by Payment Method
- Churn by Gender
- Churn by Tech Support
- Churn by Online Security
- Churn by Tenure Group

Dashboard 3

Revenue & Customer Insights

Analyse revenue contribution and customer value.

KPIs

- Total Revenue
- Average Revenue per Customer
- Highest Revenue Contract

- Highest Revenue Payment Method

Charts

- Revenue by Contract
- Revenue by Internet Service
- Revenue by Payment Method
- Average Monthly Charges by Contract
- Average Tenure by Contract
- Revenue by Tenure Group

12. DAX Measures Created

Examples:

- Total Customers
- Active Customers
- Churn Customers
- Churn Rate
- Total Revenue
- Average Revenue
- Average Monthly Charges
- Average Tenure
- Highest Revenue Contract
- Highest Revenue Payment Method

13. Data Visualization

Different visualization techniques were used.

- KPI Cards
- Clustered Column Charts
- Clustered Bar Charts
- Donut Charts
- Slicers
- Cards
- Navigation Buttons
- Interactive Filters

14. Dashboard Features

- Interactive Slicers
- Drill-down Analysis
- Dynamic KPIs
- Responsive Layout
- Business Insights Panel
- Consistent Colour Theme
- Cross Filtering
- Navigation Between Reports

15. Storytelling

The project tells a complete business story through three dashboards:

Executive Overview

Provides a high-level snapshot of customers, churn, revenue, and key performance indicators to understand the overall health of the business.

Customer Churn Analysis

Identifies why customers leave by analysing churn across contract types, internet services, payment methods, tenure, and support services. This helps the business recognize high-risk customer groups and plan targeted retention strategies.

Revenue & Customer Insights

Focuses on revenue generation by examining contract types, payment methods, customer tenure, and service categories. It highlights the most valuable customer segments and supports decisions to maximize long-term profitability.

Together, these dashboards guide stakeholders from **"What is happening?"** to **"Why is it happening?"** and finally **"Where should the business focus next?"**

16. Key Business Insights

- Approximately **26.5%** of customers have churned.
- Month-to-month contract customers have the highest churn rate.
- Fiber optic users contribute significantly to churn.

- Electronic check is the most frequently used payment method.
- Customers with shorter tenure are more likely to leave.
- Two-year contracts generate the highest revenue.
- Long-term customers contribute more revenue than new customers.
- Average monthly charges are higher among churned customers.

17. Business Recommendations

- Encourage customers to switch to long-term contracts.
- Improve customer support for new customers.
- Offer loyalty programs for high-risk customers.
- Introduce personalized retention offers.
- Improve technical support services.
- Provide discounts for month-to-month customers transitioning to yearly plans.
- Monitor customers with high monthly charges and low tenure.

18. Project Outcome

This project successfully transformed raw telecom customer data into meaningful business insights using Power BI, PostgreSQL, Python, Pandas, and Excel.

The dashboards enable business users to:

- Monitor customer churn.
- Analyse revenue performance.
- Identify valuable customer segments.
- Make informed, data-driven decisions.
- Develop strategies to reduce churn and improve customer retention.

19. Skills Demonstrated

Technical Skills

- Power BI
- DAX
- PostgreSQL
- SQL

- Python
- Pandas
- Excel
- Data Cleaning
- Data Transformation
- Data Modelling

Analytical Skills

- KPI Design
- Customer Segmentation
- Churn Analysis
- Revenue Analysis
- Dashboard Development
- Business Intelligence
- Data Visualization
- Storytelling with Data

20. Future Enhancements

- Develop a machine learning model to predict customer churn.
- Integrate real-time data refresh.
- Connect directly to cloud databases.
- Add forecasting and trend analysis.
- Implement Row-Level Security (RLS) for different user roles.
- Publish the dashboard to the Power BI Service for enterprise sharing.

21. Conclusion

The **Telco Customer Churn Analysis** project demonstrates the complete data analytics lifecycle—from data cleaning and preprocessing to SQL analysis, interactive dashboard creation, and business storytelling. By leveraging **Excel, Python (Pandas), PostgreSQL, and Power BI**, the project converts raw customer data into actionable insights that support customer retention strategies, revenue optimization, and informed decision-making. It also showcases practical skills in data preparation, DAX, visualization, and business intelligence, making it a strong portfolio project for a Data Analyst role.